Davidson Fellows Scholarship

Stress Shift

A Real-Time Embedded Control System Utilizing Inverse Dynamics for Adaptive Knee Torque Redistribution

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Abstract

Knee pain affects nearly one quarter of adults, severely limiting mobility and quality of life. Consequently, there is a growing need for preventative technologies that reduce joint stress before surgical intervention becomes necessary. Existing knee braces provide general, untargeted support that cannot respond to uneven loads. This project developed Stress Shift, a dynamically adaptive knee brace designed to redistribute harmful joint loading due to external ground forces by applying mechanical compensation during high-stress phases of gait. Pressure sensors at the heel and toe and inertial measurement units above and below the knee were used to collect real - time gait data from three able - bodied participants (ages 18-65; mean height 1.77 m) during repeated walking trials. These signals were used to estimate ground reaction forces and joint angles in real time, which were then processed through an inverse dynamics model to compute sagittal-plane external knee joint torque, identify peak stress windows, and generate subject-specific engagement thresholds. These thresholds power the variable-stiffness knee brace, in which actuators, dampers, and micro-servos were engineered to provide mechanical support during high-load gait phases. Compared to the standard brace, the adaptive brace reduced peak sagittal-plane knee torque by at least 20% across repeated walking trials. By combining biomechanics, sensing, and adaptive mechanical design, Stress Shift aims to provide a more effective and personalized approach to knee rehabilitation and injury prevention, revolutionizing the fields of orthopedics and physical therapy. Future work includes internal force compensation, a diversified dataset, and multi-actuator integration for more specific and targeted treatment.

Introduction

Background

Knee pain affects nearly 25% of adults, and knee replacement surgeries in the United States have increased eightfold over the past four decades (Li et al., 2025). Lower-limb conditions significantly impair mobility and quality of life. Moreover, surgical intervention carries substantial risks, including infection, prosthetic revision, postoperative stiffness, and persistent pain (Nguyen et al., 2011). In addition, rehabilitation following knee injury or surgery is often difficult and physically demanding, with common side effects including pain, swelling, stiffness, muscle weakness, and loss of independence (Island Hospital, 2025). To address these issues, orthopedic physicians and physical therapists have increasingly incorporated supportive rehabilitation technologies into their treatment plans. One of the most common approaches is compression wrapping or knee sleeves, which provide general support, reduce swelling, and improve stability (The Orthopedic Clinic, 2025). However, conventional knee sleeves and orthopedic braces offer uniform, static support and cannot provide targeted assistance for complex or evolving injuries. This lack of adaptability limits their ability to respond to changes in gait, load distribution, or fatigue over time, substantially slowing recovery.

Literature Review

Current work on adaptive knee bracing tends to fall into one of two categories. Some sleeves are simply custom designed to better fit the user. This method of recovery device development can be far more expensive and inaccessible than its worth, as a better fit on a conventional brace can only provide marginal improvement to adaptability. Other sleeves focus on adaptability through strategic material usage and development. These sleeves use materials

with varying stiffness or elasticity to provide passive, adaptive compression. However, this adaptation is still generalized and not tailored to the individual user, and they are extremely costly. Research shows mixed functional outcomes and no consensus that these generalized adaptations significantly improve knee function in all users (Mohd et al 2017).

Engineering Goal

Can a dynamically adaptive knee sleeve be designed to redistribute loads, reducing elevated knee torque peaks by 20% at critical gait phases while remaining low-cost, accessible, and suitable for everyday use?

Solution

The solution is Stress Shift, a novel advancement in adaptive bracing that uses dynamic load redistribution to reduce joint stress. Stress Shift is designed to stiffen only under harmful mechanical loads in specific areas while remaining flexible during normal motion. Rather than uniformly constraining the joint, the system selectively intervenes when abnormal stress patterns are detected. Stress Shift is distinct from current knee sleeves on the market: instead of relying on passive load redistribution through flexible or compressive materials, it operates as an actively adaptive system that integrates real-time sensors, a personalized processing unit, and mechanically responsive actuators to modulate support based on the user's biomechanics. This approach enables targeted, responsive assistance that evolves with movement and loading conditions rather than providing static, generalized compression.

Performance testing demonstrated that Stress Shift reduced peak sagittal-plane knee torque by 32.09% compared to standard braces, exceeding the 20% design target. The device was also engineered for accessibility and practicality: the total cost of construction was

approximately $150, representing a 62.5% reduction compared to comparable adaptive sleeves on the market starting at $400, and all components were off-the-shelf with a single supplier per key part. Designed for everyday usability, the final prototype weighs 246 g, takes approximately 5.5 minutes to put on and 2.5 minutes to remove, and can be integrated into routine use without specialized equipment.

Methodology

Materials

- MPU 6886 6-Axis IMU (2)
- Force-sensing resistor (2)
- 25kg High-torque RC Servo (4)
- Half Breadboard (1)
- ESP32 Dev Module (1)
- 5v Battery Pack/Power Bank (1)
- Neoprene Knee Sleeve (1)
- Nylon Webbing Strap (4)
- USB Cord (2)
- Small Container with Slots (1)
- Wiring, Velcro, and Adhesive
- Python, Arduino IDE, Excel Applications

Procedure

Stress Shift was developed in three phases: initial hardware setup and data acquisition, processor development, and final hardware setup with testing.

1. *Initial Hardware Setup and Data Acquisition*

The first stage of development involved wiring a wearable data-collection device to capture gait data. An ESP32 microcontroller with Wi-Fi and Bluetooth capabilities was wired to two MPU-6886 inertial measurement units (IMUs) and two force-sensitive resistor (FSR)

pressure sensors. The IMUs were mounted above and below the knee on the thigh and shank, and the FSRs were placed at the heel and toe of the foot.

The microcontroller was programmed in Arduino IDE to record timestamped sensor data and stream it via UDP to a connected computer. Data streams included acceleration, angular velocity, and foot pressure. These data streams were saved in CSV format for later analysis.

Three voluntary participants wore the device and completed five walking trials each, with each trial lasting 50 seconds. The IMUs were aligned to the sagittal plane for consistent angular measurements, and heel and toe sensor positions were defined in fixed coordinates for later calculations.

Baseline calibration was added to reduce sensor noise and variations across trials. Relative pressure values were computed using:

$$h_{rel} = max(0, p_h - h_0)$$

$$t_{rel} = max(0, p_t - t_0)$$

Median filtering was also applied to reduce noise and stabilize pressure. These safeguards resulted in cleaner, synchronized kinematic and pressure information across multiple gait cycles for each participant.

2. Processing Program Development

A Python-based processing program was developed to convert gait data into indicators of knee torque, which indicates when assistance should be applied.

A. IMU-Based Kinematic Computation

Segment orientations for the thigh and shank were calculated using a complementary filter that combined gyroscope and accelerometer data:

$$\theta(t) = \alpha \times (\theta(t - dt) + \omega \times dt) + (1 - \alpha) \times \theta_{acc}$$

$$\alpha = 0.98$$

From these segment orientations, knee angle was computed as the relative rotation between the shank and thigh:

$$\theta_{knee} = \theta_{shank} - \theta_{thigh}$$

Angular velocity and angular acceleration of the knee were calculated through differentiation and low-pass filtering of the IMU signals.

B. Ground Reaction Force and Stance Detection

Heel and toe pressure sensor data were converted into vertical ground reaction force (GRF). After scaling to body weight, force scaling during stance phase was determined using:

$$F_z = \frac{h_{rel} \times (mg)}{mean(rel_{sum})} + \frac{t_{rel} \times (mg)}{mean(rel_{sum})}$$

Center of pressure (CoP) along the foot was then calculated:

$$x_{cop} = \frac{F_h \times x_h \times F_t \times x_t}{F_h + F_t}$$

Stance phase detection was defined by vertical force thresholds:

Stance ON when:

$$F_z > 0.15 \times mg$$

Stance OFF when:

$$F_z < 0.08 \times mg$$

These calculations enabled continuous tracking of loading distribution during walking.

C. Knee Torque Computation

A simplified sagittal-plane biomechanical model was used to calculate knee joint torque. Anthropometric relationships for the shank were defined as:

$$m_2 = 0.0465 \times m$$

$$a_2 = 0.43 \times l_2$$

$$I_2 = m_2 \times (0.30 \times l_2)^2$$

The torque proxy was then calculated:

$$\tau_{\text{knee}} = I_2\ddot{\theta} + m_2 g a_2 \sin(\theta_{\text{knee}}) - F_z x_{\text{cop}}$$

Computed torque values were smoothed using a moving average filter to produce a more stable signal.

D. Control Logic

Baseline torque levels for each participant were learned during normal walking. Separate medians were computed for extension and flexion torque:

$$base_{pos} = \text{median}(\tau_f \mid \tau_f > 0)$$

$$base_{neg} = \text{median}(\{|\tau_f| : \tau_f < 0\})$$

Engagement thresholds were defined as:

- Activation when torque exceeded 1.6 times the baseline
- Deactivation when torque dropped below 1.3 times the baseline

Assist levels were scaled proportionally to torque magnitude:

$$level = \frac{tau_f - 1.6 \times base_{pos}}{1.6 \times base_{pos}}$$

$$level = \frac{|tau_f| - 1.6 \times base_{neg}}{1.6 \times base_{neg}}$$

If torque exceeded safe thresholds, the processor generated commands for mechanical assistance. High extension torque triggered flexion assistance, while high flexion torque triggered extension assistance. Commands were transmitted in real time to the ESP32 via UDP as NEUTRAL, FLEX, or EXT, each scaled from 0 to 1.

3. Final Hardware Integration

To channel the Python processor's outputs into mechanical assistance, a knee sleeve was modified using four servo motors. Two servos were mounted on the thigh and two on the shank. Each servo used a spool attachment connected to nylon straps integrated into the sleeve.

When activated, the servos tightened or loosened these straps to apply controlled assistance across the knee joint. All four servos were wired to the ESP32 microcontroller through GPIO pins and controlled by UDP commands from the Python processor.

The ESP32 was programmed to interpret processor commands as follows:

- NEUTRAL: No servo movement
- FLEX: Thigh servos tightened while shank servos loosened
- EXT: Shank servos tightened while thigh servos loosened

The magnitude of servo rotation was proportional to the assistance level calculated by the processor. Rate limiting and timeout functions were for safe operation.

4. Testing and Evaluation

The system was evaluated through walking trials with the same three participants from before. Each participant completed two sets of trials:

- Standard trials: Walking with the sleeve worn but adaptive servo assistance disabled
- Adaptive trials: Walking with real-time servo assistance enabled

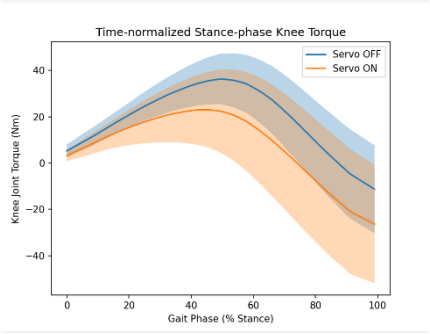
For each trial, processed data including ground reaction force, torque, filtered torque, and stance phase timing were recorded and analyzed.

Percentage improvements were calculated using percent reduction in peak knee torque:

$$\% \text{ Reduction} = \frac{\tau peak_{standard} - \tau peak_{adaptive}}{\tau peak_{standard}} \times 100$$

Torque and peak plots were used to determine whether Stress Shift reduced peak loading during high-stress phases of gait.

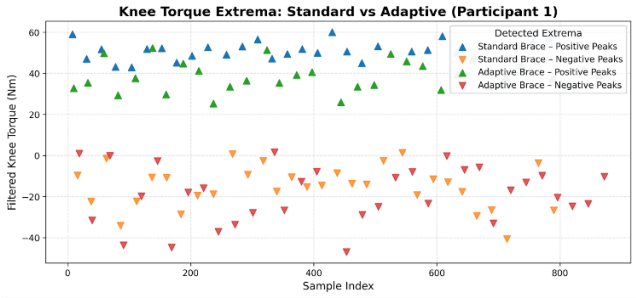
Results

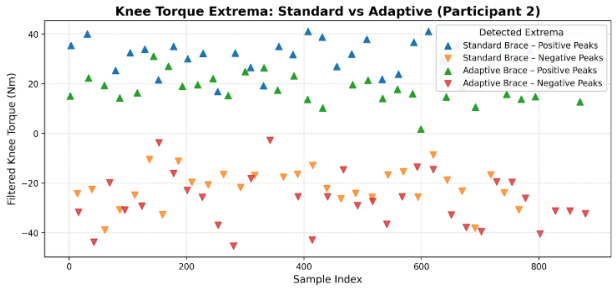


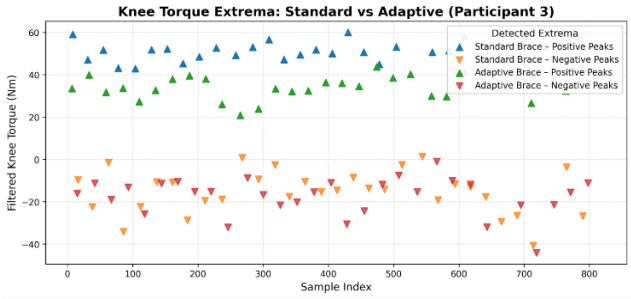
Figures 1 and 2: Average torque per participant was computed for both the standard and adaptive sleeve.

Mode	P1 (Nm)	P2 (Nm)	P3 (Nm)	Avg
Standard	10.625	4.012	12.648	9.095
Adaptive	5.217	-1.804	6.415	3.276

Average knee torque across the stance phase decreased for all participants when using the adaptive sleeve compared to the standard sleeve. The standard sleeve produced an average torque of 9.095 Nm across participants, while the adaptive sleeve produced 3.276 Nm. This is an overall average reduction of 32.09% in sagittal-plane knee torque.







Peak Knee Torque Comparison: Standard vs. Adaptive Brace

Participan	Brace	Pos Count	Avg Pos Peak (Nm)	Neg Count	Avg Neg Peak (Nm)
1	Standard	31	50.28	31	-15.28
	Adaptive	**33**	**38.78**	**33**	**-19.04**
2	Standard	31	31.09	30	-21.87
	Adaptive	**33**	**18.06**	**33**	**-27.12**
3	Standard	31	50.28	31	-15.28
	Adaptive	**31**	**34.44**	**30**	**-17.38**

Figures 3 and 4: Average knee torque extrema for all three subjects.

Reductions in positive torque peaks were observed consistently across all three participants. Participant 1 demonstrated a 22.87% reduction, Participant 2 a 41.91% reduction, and Participant 3 a 31.5% reduction, resulting in a cumulative average decrease of 32.09%. Negative torque peaks also decreased, though to a lesser extent, with an average reduction of

19.05%. These findings indicate that the adaptive sleeve reduced stressful loading during key phases of the gait cycle.

Subject	Flex	Ext
P1	57	19
P2	48	21
P3	56	20

Figure 5: Average flexion and extension trigger counts.

Flexion and extension trigger counts remained relatively consistent across participants, with slightly higher flexion-phase activations. This consistency suggests stable control behavior of the system across users.

Conclusion

The final Stress Shift prototype consisted of a wearable knee sleeve with real-time sensing, biomechanical processing, and adaptive mechanical assistance. Stress Shift reduced peak sagittal-plane knee torque by 32.09% compared to standard braces, exceeding the 20%

target. The impacts of an effective, adaptive treatment system are profound and revolutionary. When a device adapts with the user's recovery and provides specific, targeted support, users must expend less time, effort, and money to achieve the results they want. Stress Shift has widespread applications across various fields and different injury types.

Stress Shift blends the accessibility of the traditional knee sleeve with the advanced technological prowess of clinical treatments, allowing patients to bring high-tech treatments into the comfort of their homes.

- Cost: Stress Shift comes in at $150; a 62.5% reduction compared to similar adaptive sleeves on the market starting at $400.
- Accessibility: All parts were easily accessible for purchase, with 1 supplier per key component and only off-the-shelf materials.
- Everyday Use: The finished product took approx. 5.5 minutes to put on and 2.5 minutes to remove, with a weight of 246g.

There were several constraints that influenced the development of this project. I did not have access to a specialized biomechanics lab or advanced testing equipment, so all aspects of development were completed using off-the-shelf tools and materials. I also completed the development of Stress shift without a mentor or expert in biomechanics to guide the process, so I relied on publicly available research and independent, iterative testing. Finally, without outside funding or a grant, the project had to remain within a limited budget. These constraints shaped many design choices and demonstrated that an adaptive, orthopedic device can still be developed using accessible resources.

While the current prototype demonstrated strong performance, it also formed the basis for several areas of future development.

- Passive Torque Compensation: The current model did not address passive torques that occur within the joint, only torques that are applied to the joint from the ground. Passive torques can be estimated using IMUs positioned closer to the knee, along with more complex computational physics, for a more accurate, effective device.
- Diversified Dataset: My project relied on the data and testing of 3 anonymous participants. For a more scalable product, my dataset could be expanded to include more participants with varied physical demographics.
- Multi-Actuator Integration: The current model uses four servos to provide active torque redistribution around the joint. Future models could experiment with more subtle, targeted actuators such as shape memory alloy and mini linear, motor-based actuators.

Works Cited

Li, Y., Ajia, A., Wu, Z., Fan, J., Wu, M., Mao, P., Yang, F., He, Q., & Wang, P. (2025). Previous knee surgery increases risks of revision, infection, pain and stiffness after knee replacement

arthroplasty: a systematic review and meta-analysis. BMC musculoskeletal disorders, 26(1), 869. https://doi.org/10.1186/s12891-025-09060-6

Limb Reconstruction Surgery Recovery - Island Hospital. (2025, July 14). Island Hospital. https://islandhospital.com/limb-reconstruction-surgery-recovery/

Mohd Sharif, N. A., Goh, S. L., Usman, J., & Wan Safwani, W. K. Z. (2017). Biomechanical and functional efficacy of knee sleeves: A literature review. Physical therapy in sport : official journal of the Association of Chartered Physiotherapists in Sports Medicine, 28, 44–52. https://doi.org/10.1016/j.ptsp.2017.05.001

Nguyen, U. S., Zhang, Y., Zhu, Y., Niu, J., Zhang, B., & Felson, D. T. (2011). Increasing prevalence of knee pain and symptomatic knee osteoarthritis: survey and cohort data. Annals of internal medicine, 155(11), 725–732. https://doi.org/10.7326/0003-4819-155-11-201112060-00004

The Orthopedic Clinic. (2025, March 3). 15 Home Remedies for Knee Pain - The Orthopedic Clinic. The Orthopedic Clinic. https://orthotoc.com/home-remedies-for-knee-pain/

Kondratiev, A., Smetankina, N., & Staude, V. (2025). Biomechanical analysis of stress–strain

distribution in the lumbar spine–sacrum–pelvis system with emphasis on sacroiliac joint dysfunction. Prosthesis, 7(1), Article 4. https://doi.org/10.3390/prosthesis7010004

Ou, D., Ye, Y., Pan, J., Huang, Y., Kuang, H., Tang, S., Huang, R., Mo, Y., & Pan, S. (2024). Finite element study of stress distribution in medial UKA under varied lower limb alignment. Scientific Reports, 14(1), Article 74145. https://doi.org/10.1038/s41598-024-74145-6

Li, Z., Xu, G., Wang, C., Wang, Q., Liu, C., Guo, T., Wu, L., & Cao, D. (2023). Variation characteristics of stress distribution in the subchondral bone of the knee joint of judo athletes with long-term stress changes. Frontiers in Endocrinology, 13, Article 1082799. https://doi.org/10.3389/fendo.2022.1082799

Boudjemaa, I., Albedah, A., Sahli, A., Bouiadjra, B. A., Benkhettou, A., & Benbarek, S. (2024). Innovative stress-responsive liner design for lower limb disability. Journal of Disability Research, 3(8). https://doi.org/10.57197/jdr2024-0098

Koshio, T., Haraguchi, N., Takahashi, T., Hara, Y., & Hase, K. (2024). Estimation of ground reaction forces during sports movements by sensor fusion from inertial measurement units with a 3D forward dynamics model. Sensors, 24(9), Article 2706. https://doi.org/10.3390/s24092706